

4.7 Change of Variables in Multiple Integrals Worksheet

1. Let R be the region bounded by the curves $x^2 + y^2 = 1$, $y = \frac{1}{2}$, & $y = -\frac{1}{2}$. Let $v = y$ and $u = \frac{x}{\sqrt{1-y^2}}$.

a) Solve the equations for x as a function of u and v . Similarly, obtain y as a function of u and v . That is, determine $x(u, v)$ & $y(u, v)$.

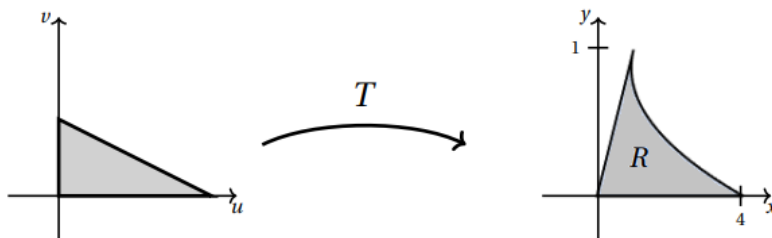
b) Let $v = y$ and $u = \frac{x}{\sqrt{1-y^2}}$. Draw the region R in the u, v coordinate system.

c) What is $\left| \frac{\partial(x,y)}{\partial(u,v)} \right|$?

d) Use your answers from parts (a) – (c) to perform a change of variables on the double integral

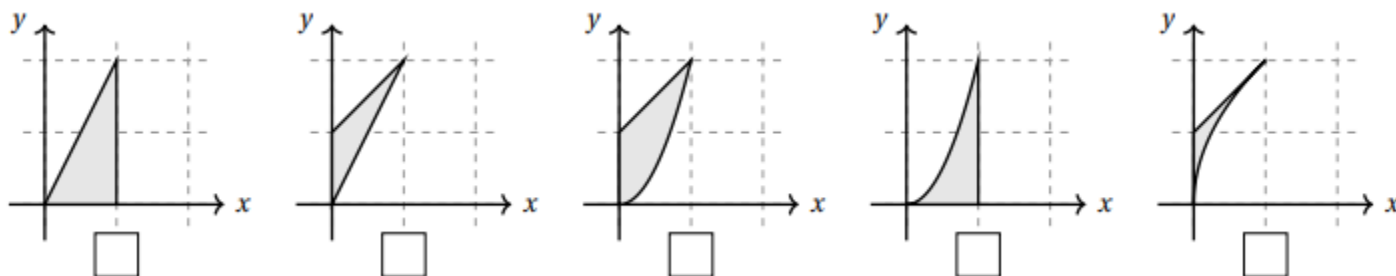
$$\iint_R \frac{1}{\sqrt{1-y^2}} dy dx.$$

2. The transformation $T(u,v) = (u^2 + v, v)$ takes the triangle with vertices $(0,0)$, $(2,0)$, & $(0,1)$ to the region R as shown below. Evaluate *area of* R .



3. Let T be the transformation given by $T(u, v) = (uv, u + v)$. Let S be the triangle in the uv -plane whose vertices are $(0, 0)$, $(1, 0)$, $(1, 1)$. Let R be the region obtained when T transform S into a region in the xy -plane, that is $T(S) = R$.

a) Check the box below the picture which represents the region R (assume axes are scaled by 1s).



b) Express $\iint_R y dA$ as an integral over S .

4. For each transformation T below circle “yes” or “no” depending on whether or not it takes the rectangular region S , $0 \leq u \leq 1$, $0 \leq v \leq 2$ in the uv -plane to the parallelogram region R in the xy -plane with vertices $(0,0)$, $(4,2)$, $(2,-2)$, & $(6,0)$.

a) YES NO $T_1(u,v) = (2u + 4v, -2u + 2v)$

b) YES NO $T_2(u,v) = (4u + v, 2u - v)$

c) YES NO $T_3(u,v) = (4u + 2v, 2u - 2v)$

d) YES NO $T_4(u,v) = (2u + 2v, -2u + v)$

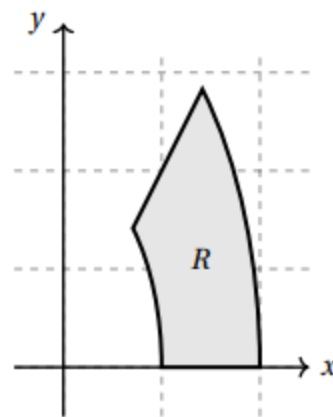
5. Let R be the rectangle whose vertices are $(1,0)$, $(2,1)$, $(3,-2)$, & $(4,-1)$ shown at the right.
- a) Exactly one of the following defines a transformation $T(u,v)$ from the uv -plane to the xy -plane with $T(S) = R$, where $S = \{(u,v) | 0 \leq u \leq 1, 0 \leq v \leq 1\}$. Circle the correct formula for $T(u,v)$.
- | | |
|--------------------------------------|--------------------------------------|
| i) $T(u,v) = (2u + 3v, u - 2v)$ | ii) $T(u,v) = (2u + 4v, u - v)$ |
| iii) $T(u,v) = (2u + 4v + 1, u - v)$ | iv) $T(u,v) = (2u + 3v + 1, u - 2v)$ |
| v) $T(u,v) = (u + 2v, u - 2v)$ | vi) $T(u,v) = (u + 2v + 1, u - 2v)$ |

- b) Determine if $\iint_R y \, dA$ is negative, zero, or positive.

6. Let E be the trapezoid with vertices $(8,0)$, $(12,0)$, $(0,8)$, and $(0,12)$. Using the change of variables $u = x - y$ & $v = x + y$ change $\iint_E (x^2 - y^2) dA$ into an integral that is easier to evaluate. You do not need to evaluate the resulting integral.

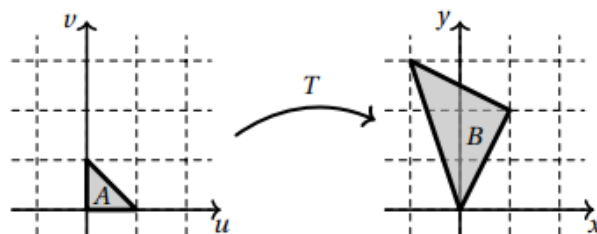
7. Use the transformation $x = u\sqrt{2} - v\sqrt{\frac{2}{3}}$ & $y = u\sqrt{2} + v\sqrt{\frac{2}{3}}$ to evaluate the integral $\iint_R (x^2 - xy + y^2) dA$, where R is the region bounded by the ellipse $x^2 - xy + y^2 = 2$.

8. Suppose R is the region in the first quadrant between the ellipses $x^2 + \frac{y^2}{4} = 1$ and $x^2 + \frac{y^2}{4} = 4$ and the lines $y = 0$ and $y = 2x$ shown at the right. Using the transformation $T(u, v) = (u \cos(v), 2u \sin(v))$, find the integrand and limits of integration expressing the integral $\iint_R x \, dA$ as an iterated integral over a subset S in the uv -plane with $T(S) = R$.



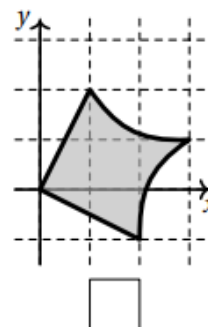
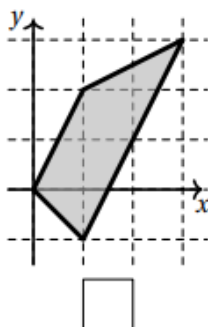
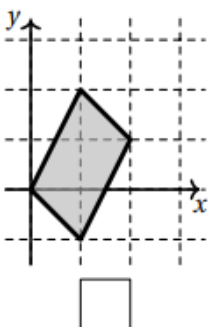
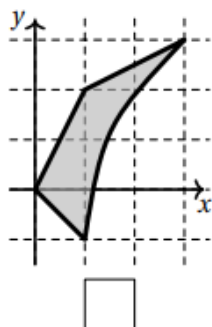
9. Find a transformation T that transforms the triangular region with vertices $(0,0)$, $(0,1)$, $(1,0)$ to the triangular region B with vertices $(0,0)$, $(1,2)$, $(-1,3)$.
As shown in the picture to the right.

We will try to create a transformation which maps
 $(0,0) \rightarrow (0,0)$, $(0,1) \rightarrow (1,2)$, and $(1,0) \rightarrow (-1,3)$.



10. Let T be the transformation $T(u,v) = (u+v+uv, -u+2v+2uv)$. Let $S = \{(u,v) | 0 \leq u \leq 1 \text{ \& } 0 \leq v \leq 1\}$ be the unit square in the uv -plan and let R be the region that is the image of S under T .

a) Check the box below that corresponds to the region R .



- b) Fill in the limits and integrand of the integral below so that it computes $\iint_R \sqrt{x} \, dA$ as an integral over the square S .

$$\iint_R \sqrt{x} \, dA = \int \int \quad \quad \quad dudv$$

11. Use an appropriate change of variables to evaluate the integral $\iint_R \frac{x-y}{2x+2y} dA$, where R is the rectangle with vertices $(1,0)$, $(2,-1)$, $(3,2)$, $(4,1)$.

12. Let E be the region bounded by $16x^2 + \frac{y^2}{9} = 2$. Find an appropriate change of variables and change

$\iint_E e^{\left(16x^2 + \frac{y^2}{9}\right)} dA$ into an integral that is easier to evaluate. You do not need to evaluate the resulting integral.

13. Let R be a parallelogram with vertices $(1,0)$, $(3,1)$, $(2,3)$, $(0,2)$. If $f(x,y) = (2y-x)^2(y+2x)$, then compute $\iint_R f(x,y) dA$.